

Hybrid Co-Simulation Algorithm

Master

Event
Sources

Listeners

Macro-Step [t, t+dt]

t_left = t, t_right = t+dt

loop [while t_left < t_right]

1. Prepare inputs at t_left

ready

2. Detect events [t_left, t_right]

snapshot → trial step → detect crossings → restore

crossings found?

alt [No events]

Gauss-Seidel step
[t_left, t_right]

t_left = t_right
EXIT LOOP

[Events detected]

3. Bisection:
Locate t_event

4. Step all
[t_left, t_event]

step to t_event

loop [5. Superdense time (t_event, n)]

Handle events

handle_events_internal()

done

Detect new events?

alt [New events]

n ← n+1

[No new events]

Exit microstep loop

t_left ← t_event

Continue loop

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